

# Determination of minority carrier diffusion length and back-surface recombination velocity in thin-film solar cells using lateral photocurrent

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Eunyeong Yang,<sup>1</sup> Jiseong Jang,<sup>1</sup> Juhyeon Song,<sup>2</sup> Jung-Sub Wi,<sup>2</sup> Joon Sik Park,<sup>2</sup> Sangyeob Lee,<sup>2,a)</sup> and Choong-Heui Chung<sup>1,2,a)</sup>

## AFFILIATIONS

<sup>1</sup>Department of Materials and Manufacturing Engineering, Hanbat National University, Daejeon 34158, Republic of Korea

<sup>2</sup>Department of Materials Science and Engineering, Hanbat National University, Daejeon 34158, Republic of Korea

<sup>a)</sup>Authors to whom correspondence should be addressed: [sangyeob@hanbat.ac.kr](mailto:sangyeob@hanbat.ac.kr) and [choong@hanbat.ac.kr](mailto:choong@hanbat.ac.kr)

## ABSTRACT

Accurate determination of the minority carrier diffusion length ( $L_d$ ) and back-surface recombination velocity ( $S_b$ ) is essential for understanding carrier transport and recombination in thin-film solar cells, yet remains challenging due to the concurrent effects of drift-diffusion transport and multiple recombination pathways. In this work, we present a method to extract  $L_d$  and  $S_b$  from wavelength-dependent lateral photocurrent measurements. The approach exploits the anisotropy of carrier transport in bandgap-graded absorbers, where the quasi-electric field is predominantly oriented along the thickness direction, resulting in diffusion-dominated lateral transport. By relating the lateral collection length to an effective diffusional collection length through two limiting cases, bounded estimates of  $L_d$  and  $S_b$  are obtained without requiring explicit knowledge of the bandgap profile. The method is applied to a Cu(In,Ga)Se<sub>2</sub> thin-film solar cell, yielding  $L_d \approx 1.8 \mu\text{m}$  and  $S_b$  in the range of  $5.3 \times 10^4$ – $1.2 \times 10^5$  cm/s, consistent with reported values. The proposed framework provides a physically motivated approach to separate transport and recombination contributions using experimentally accessible quantities and is applicable to both graded and ungraded absorber systems.

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## I. INTRODUCTION

Accurate characterization of carrier transport and recombination is essential for understanding and optimizing thin-film solar cells. Recombination in a solar cell can be classified into front-interface, depletion region, bulk, and back-surface pathways.<sup>1</sup> Among them, two key parameters primarily govern solar cell quality: the minority carrier diffusion length ( $L_d$ ), which characterizes bulk transport, and the back-surface recombination velocity ( $S_b$ ), which quantifies recombination losses at the back-surface.<sup>2–6</sup> Independent determination of these parameters is challenging for two reasons.

First, bulk recombination and back-surface recombination occur concurrently, so any measured response reflects their combined effect. Extracting  $L_d$  and  $S_b$ , therefore, requires decoupling these two loss mechanisms. Second, carrier transport involves simultaneous diffusion and drift. Because  $L_d$  is defined as a purely

diffusional quantity, the drift contribution must also be separated from the measured transport. Drift arises from two distinct electric fields. In the space-charge region (SCR), the built-in field of the p–n junction drives carriers by drift. In the charge-neutral region (CNR), a quasi-electric field can develop when the bandgap of the absorber is graded along the thickness direction—a design strategy commonly employed to enhance charge-collection efficiency.<sup>7–13</sup>

The quantum efficiency method is widely used to measure the collection length  $L_c$ .<sup>14,15</sup> However,  $L_c$  is an effective parameter that conflates diffusion with drift and bulk recombination with back-surface recombination; it does not isolate any single transport or recombination mechanism. Bias-dependent quantum efficiency measurements have been proposed to decouple recombination processes by modulating the SCR width,<sup>16</sup> but this approach still cannot distinguish diffusion from drift in the CNR. Its applicability is, therefore, limited in solar cells with bandgap grading. Because the quasi-electric field in the CNR introduces a drift

contribution that is superimposed on diffusion, any measurement of the collection length will reflect their combined effect rather than the diffusional component alone.

Here, we present a method that extracts  $L_d$  and  $S_b$  from experimentally measurable quantities. Rather than fully resolving drift-diffusion transport, the approach isolates the dominant diffusional contribution through a geometrical decomposition of carrier motion and provides bounded estimates of the relevant parameters. The method is demonstrated on a Cu(In,Ga)Se<sub>2</sub> (CIGS) thin-film solar cell and is applicable to both ungraded and graded-bandgap absorbers without requiring explicit knowledge of the bandgap profile.

## II. FRAMEWORK FOR EXTRACTING $L_d$ AND $S_b$

To overcome the limitations described above, a custom-designed structure is introduced (Fig. 1). The structure consists of a bottom electrode, a light absorber, and an opaque top electrode that serves as a collector of photogenerated minority carriers. Monochromatic light illuminates the structure outside the periphery of the opaque electrode, generating carriers over a distribution of depths governed by the exponential absorption profile. For an exponential generation profile  $G(z) = \alpha \cdot G_0 \cdot \exp(-\alpha z)$ , the quantity  $1/\alpha$  corresponds to the mean of the distribution,  $\langle z \rangle = 1/\alpha$ . This average generation depth is used as a representative measure of the generation position throughout this work. The generated carriers travel laterally toward the collector through a combination of diffusion and drift while undergoing bulk and back-surface recombination.

Before presenting the detailed derivation, we summarize the physical meaning of the four key quantities and the transport and recombination processes each reflects.  $L_{lc}$  is the directly measured lateral collection length, which contains the combined effects of drift, diffusion, and both bulk and back-surface recombination; the diffusional collection length  $L_{dc}$  isolates the diffusional contribution by removing the drift component; the minority carrier

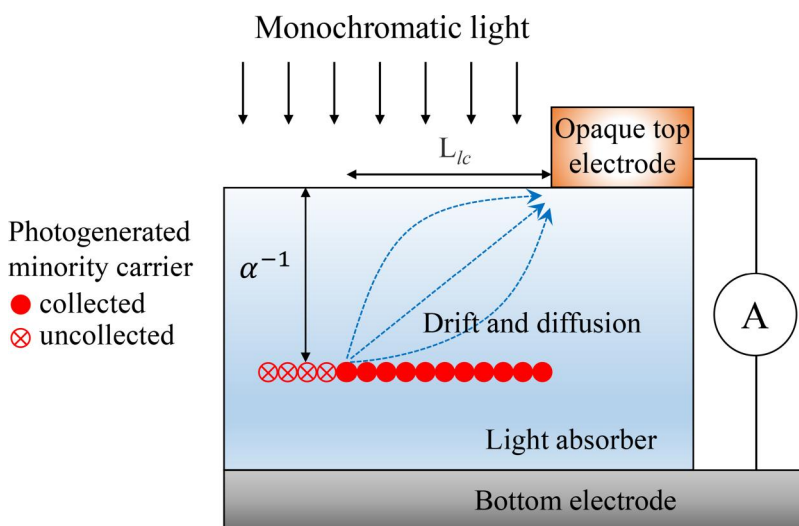
diffusion length  $L_d$  is obtained from the saturation of  $L_{dc}$  where back-surface recombination is negligible and thus reflects bulk recombination alone; and the back-surface recombination velocity  $S_b$  is extracted from the deviation of  $L_{dc}$  from  $L_d$ . The determination of  $L_d$  and  $S_b$  proceeds in three steps: (1) the lateral collection length  $L_{lc}$  is obtained from the measured lateral photocurrent; (2) the diffusional collection length  $L_{dc}$  is extracted by bounding the drift contribution under two limiting cases; and (3)  $L_d$  and  $S_b$  are determined from the depth-dependent behavior of  $L_{dc}$ .

### A. Step 1: Determination of lateral collection length ( $L_{lc}$ )

The lateral collection length  $L_{lc}$  is the characteristic lateral distance from the collector edge over which photogenerated carriers contribute effectively to the photocurrent (Fig. 1). Because the average generation depth depends on wavelength,  $L_{lc}$  also varies with wavelength: at longer wavelengths, carriers are generated deeper—closer to the back-surface—and are more strongly affected by back-surface recombination, yielding a smaller  $L_{lc}$ ; at shorter wavelengths, carriers are generated closer to the front, yielding a larger  $L_{lc}$ . Varying the wavelength, therefore, provides a depth-dependent profile of  $L_{lc}$ .

It should be noted that  $L_{lc}$  should not be interpreted as representing a single depth-resolved collection length, because carriers are generated over a range of depths and their depth coordinates can further evolve during transport. The quantity, therefore, represents an average over the generation depth rather than a value at a fixed position. This approximation is known to introduce an uncertainty of approximately 10% while still providing a reasonable estimate of the depth-dependent behavior.<sup>17</sup>

The collection process can be approximated using a step-like model: carriers generated within  $L_{lc}$  are collected, while those beyond  $L_{lc}$  are not (Fig. 1). This is consistent with the conventional model in which the collection probability decays exponentially with distance such that carriers within approximately one



**FIG. 1.** Schematic of the custom-designed test structure for lateral photocurrent measurement. The structure consists of a bottom electrode, a light absorber, and an opaque top electrode. Monochromatic light is incident outside the periphery of the opaque electrode, and photogenerated carriers travel laterally toward the collector. The average generation depth  $1/\alpha$  represents the characteristic generation position.

diffusion length provide the dominant contribution.<sup>14–16</sup> Under this approximation, the lateral photocurrent per unit perimeter of the opaque electrode is  $i_{lp} \approx L_{lc} \times J_{gen}$ , where  $J_{gen} = eI_i(1 - R_{ab})/(hc/\lambda)$ . Accordingly,  $L_{lc} = (hc/\lambda)(i_{lp}/I_i)/[e(1 - R_{ab})]$ , where  $e$  is the electron charge,  $I_i$  is the incident light intensity,  $R_{ab}$  is the front-surface reflectance,  $h$  is the Planck constant,  $c$  is the speed of light, and  $\lambda$  is the wavelength.

### B. Step 2: Determination of diffusional collection length ( $L_{dc}$ )

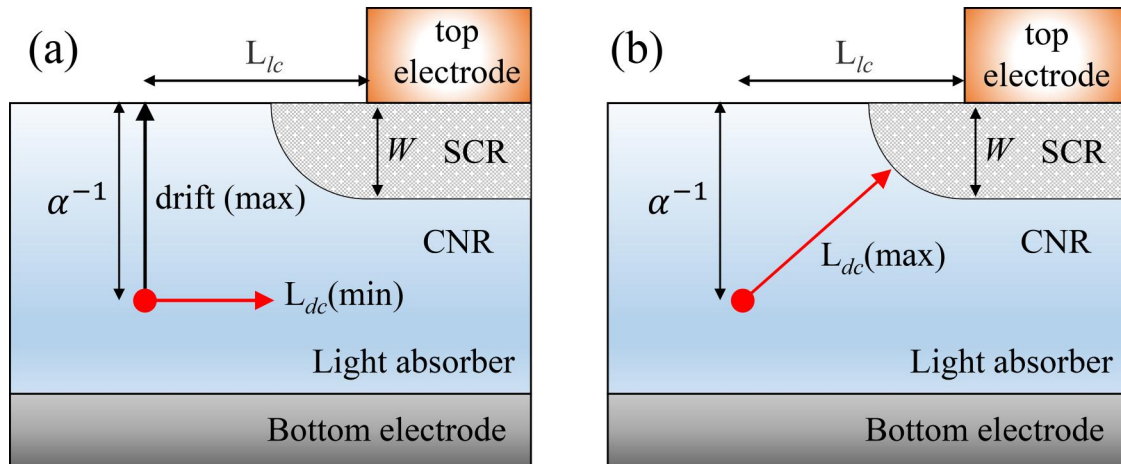
In the second step, the diffusional collection length  $L_{dc}$  is extracted from  $L_{lc}$  by exploiting the distinct transport characteristics of the CNR and the SCR. In the CNR, the quasi-electric field induced by bandgap grading is oriented predominantly along the thickness direction, and its lateral component is negligible under the assumption of lateral compositional uniformity. Under this condition, lateral carrier transport is diffusion-dominated, whereas vertical carrier displacement involves both diffusion and drift. This directional asymmetry enables isolation of the diffusional contribution from the mixed transport. As a result, the lateral transport provides a direct probe of the diffusional component independent of the drift contribution. This approximation is valid away from electrode edges and in the absence of lateral band bending or compositional gradients. In the SCR, the strong built-in field ensures near-unity collection probability, and recombination is neglected. The depletion region extends laterally at the edge of the top electrode. Following the treatment of the narrow-width effect in MOSFETs, this lateral depletion region is approximated as a quarter-circle of radius  $W$  (Fig. 2). To relate  $L_{lc}$  to  $L_{dc}$ , it is necessary to consider the carrier displacement required for photogenerated carriers to travel from their average generation position ( $1/\alpha$ ) to the SCR boundary over a lateral distance  $L_{lc}$ . This displacement can be described under two limiting cases.

Although the exact value of  $L_{dc}$  cannot be uniquely determined from a given  $L_{lc}$ , its range can be bounded by two limiting cases. Let us derive the diffusional displacement to reach the edge of the SCR region for two limiting cases. In the limit of a strong quasi-electric field in the CNR, vertical carrier displacement is dominated by drift while lateral displacement remains diffusional. The drift contribution to  $L_{lc}$  is, therefore, maximized, and the diffusional component is minimized, giving  $L_{dc}(\min) = L_{lc} - W$  [Fig. 2(a)], where  $W$  is the SCR width. In the opposite limit, where the quasi-electric field in the CNR is negligible, carrier motion in both directions is governed by diffusion. The total diffusional displacement then has both vertical and lateral components, giving  $L_{dc}(\max) = (L_{lc}^2 + \alpha^{-2})^{1/2} - W$  [Fig. 2(b)]. Applying this analysis over a range of wavelengths yields the depth-dependent profiles of  $L_{dc}(\min)$  and  $L_{dc}(\max)$ .

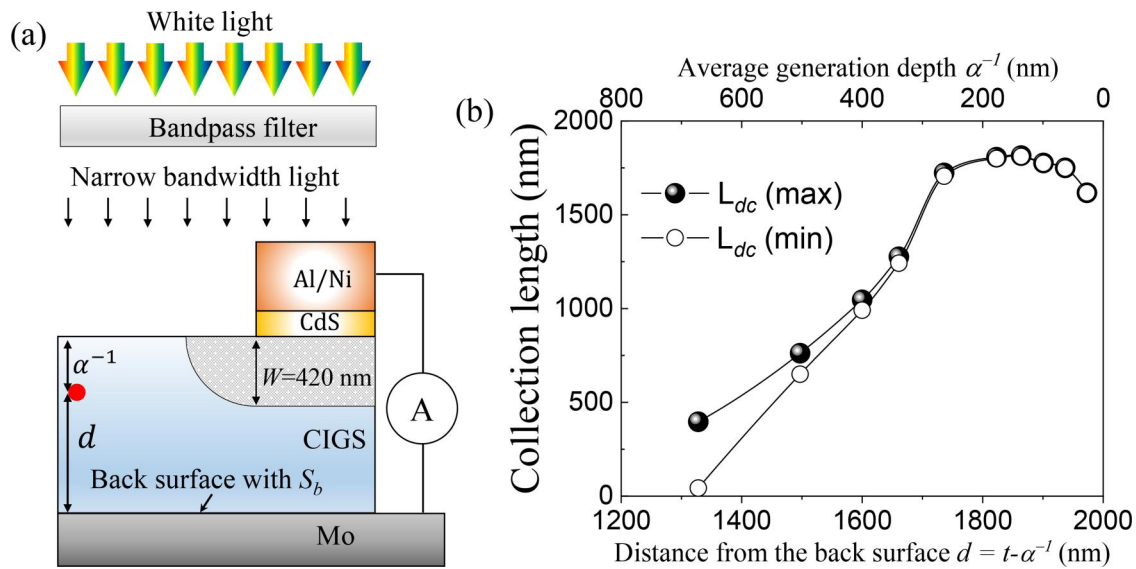
### C. Step 3: Determination of $L_d$ and $S_b$

The quantity  $L_{dc}$  isolates the diffusional contribution to transport but still reflects both bulk and back-surface recombination. Its variation with average generation depth, therefore, contains information primarily about recombination processes. As the distance between the average generation depth and the back-surface increases, the influence of back-surface recombination diminishes, and  $L_{dc}$  approaches a saturation value that represents the minority carrier diffusion length  $L_d$ .

The back-surface recombination velocity  $S_b$  is extracted from the deviation of  $L_{dc}$  from  $L_d$  in the regime where back-surface recombination remains significant. When carriers are generated at depths shallower than the SCR width ( $1/\alpha < W$ ), their displacement toward the collector is predominantly lateral, allowing the use of the following relation:  $S_b = 4dD(L_{dc}^{-2} - L_d^{-2})$ , where  $d$  is the distance from the average generation depth to the back-surface and  $D$  is the minority carrier diffusivity.<sup>18,19</sup> This condition is



**FIG. 2.** Geometrical decomposition of carrier displacement under two limiting cases for determining  $L_{dc}$  from  $L_{lc}$ . (a) Strong quasi-electric field limit: vertical displacement in the CNR is dominated by drift, and the diffusional component is minimized, giving  $L_{dc}(\min) = L_{lc} - W$ . (b) Negligible quasi-electric field limit: carrier motion in both directions is governed by diffusion, giving  $L_{dc}(\max) = (L_{lc}^2 + \alpha^{-2})^{1/2} - W$ .  $W$  denotes the space-charge region width.



**FIG. 3.** (a) Structure of the fabricated CIGS test device: Al–Ni–CdS/CIGS(2  $\mu$ m)/Mo. (b) Diffusional collection lengths  $L_{dc}(\text{min})$  and  $L_{dc}(\text{max})$  as a function of distance from the back-surface.  $L_{dc}$  saturates at 1811–1815 nm as the distance from the back-surface increases, yielding  $L_d \approx 1.8 \mu\text{m}$ .

satisfied when carriers are generated within the SCR width ( $0 < 1/\alpha < W$ ) so that their displacement is predominantly lateral rather than vertical. In this framework,  $d = t - 1/\alpha$ , where  $t$  is the absorber thickness. Because  $L_{dc}$  is bounded between  $L_{dc}(\text{min})$  and  $L_{dc}(\text{max})$ , the corresponding values of  $S_b$  are also bounded, and the actual value is expected to lie within this range.

### III. APPLICATION TO A THIN-FILM SOLAR CELL

The proposed method was applied to a CIGS thin-film solar cell with the structure shown in Fig. 3(a): Al–Ni–CdS/CIGS (2  $\mu$ m)/Mo. The CIGS layer was fabricated by a three-stage co-evaporation process, which typically produces a graded bandgap along the thickness direction. Such bandgap grading induces a quasi-electric field in the charge-neutral region (CNR), resulting in carrier transport that involves both drift and diffusion. This makes CIGS an appropriate model system for validating the present approach, which aims to isolate the diffusional component from mixed transport. In the test structure, the CdS buffer layer is confined to the region beneath the top electrode, while it is removed from the illuminated region. This is necessary because, when CdS covers the entire CIGS surface, photogenerated electrons flow preferentially along the n-type CdS layer rather than through the CIGS bulk, leading to an overestimation of the collection length. Removing CdS from the illuminated region, therefore, ensures that the measured lateral photocurrent reflects transport through the CIGS absorber itself. The SCR width  $W$  was determined to be 420 nm from capacitance measurements at 100 kHz. The incident light intensity  $I_l$  was measured with an optical power meter, and the front-surface reflectance  $R_{ab}$  was obtained by UV–vis spectrometry. The lateral photocurrent per unit perimeter  $i_p$  was recorded with a short-circuited source meter. The

absorption coefficient  $\alpha(\lambda)$  was taken from Paulson *et al.*<sup>20</sup> Table I summarizes  $\alpha^{-1}$ ,  $R_{ab}$ ,  $i_p/I_l$ ,  $L_{dc}(\text{min})$ , and  $L_{dc}(\text{max})$  at various wavelengths.

Figure 3(b) shows the extracted  $L_{dc}$  as a function of distance from the back-surface,  $d = t - 1/\alpha$ . At small  $d$ , carriers are generated close to the back-surface and are strongly affected by back-surface recombination, resulting in a reduced  $L_{dc}$ . As  $d$  increases, the influence of back-surface recombination diminishes, and  $L_{dc}$  increases and saturates at 1811–1815 nm (at average generation depth  $\alpha^{-1} = 136$  nm), indicating that back-surface recombination becomes negligible. The diffusion length is, therefore,  $L_d \approx 1.8 \mu\text{m}$ , consistent with reported values for CIGS solar cells.<sup>21–23</sup> At larger

**TABLE I.** Summary of measured and extracted parameters at various wavelengths for the fabricated CIGS test structure: average generation depth ( $\alpha^{-1}$ ), front-surface reflectance ( $R_{ab}$ ), normalized lateral photocurrent ( $i_p/I_l$ ), and bounded diffusional collection lengths [ $L_{dc}(\text{min})$  and  $L_{dc}(\text{max})$ ].

| $\lambda$<br>(nm) | $\alpha^{-1}$<br>(nm) | $R_{ab}$ | $i_p/I_l$<br>(A m/W)  | $L_{dc}(\text{min})$<br>(nm) | $L_{dc}(\text{max})$<br>(nm) |
|-------------------|-----------------------|----------|-----------------------|------------------------------|------------------------------|
| 400               | 27                    | 0.23     | $5.06 \times 10^{-7}$ | 1617                         | 1617                         |
| 500               | 63                    | 0.21     | $6.91 \times 10^{-7}$ | 1749                         | 1750                         |
| 600               | 99                    | 0.19     | $8.6 \times 10^{-7}$  | 1774                         | 1776                         |
| 700               | 136                   | 0.19     | $1.02 \times 10^{-6}$ | 1811                         | 1815                         |
| 800               | 177                   | 0.19     | $1.16 \times 10^{-6}$ | 1800                         | 1807                         |
| 900               | 264                   | 0.19     | $1.25 \times 10^{-6}$ | 1706                         | 1723                         |
| 970               | 339                   | 0.2      | $1.04 \times 10^{-6}$ | 1242                         | 1276                         |
| 1000              | 400                   | 0.18     | $9.33 \times 10^{-7}$ | 991                          | 1046                         |
| 1025              | 503                   | 0.18     | $7.25 \times 10^{-7}$ | 650                          | 762                          |
| 1050              | 672                   | 0.25     | $2.94 \times 10^{-7}$ | 43                           | 396                          |



$d$  values, where carriers are generated close to the front surface, a slight decrease in  $L_{dc}$  from the saturation value is observed, which is attributed to front-surface recombination.

To determine  $S_b$ , only  $L_{dc}$  values corresponding to generation depths shallower than the SCR width ( $1/\alpha < W$ ) are used such that carrier transport toward the collector is predominantly lateral before reaching the SCR. A measurable reduction in  $L_{dc}$  due to the back-surface recombination is also required; data at average generation depths of 339 and 400 nm meet both criteria. The diffusivity  $D$  is estimated to be  $2.57 \text{ cm}^2/\text{s}$  at room temperature using the Einstein relation with a minority carrier mobility of approximately  $100 \text{ cm}^2/\text{Vs}$ .<sup>10</sup> The resulting  $S_b$  lies in the range  $5.3 \times 10^4$ – $1.2 \times 10^5 \text{ cm/s}$ , consistent with reported CIGS values.<sup>23,24</sup>

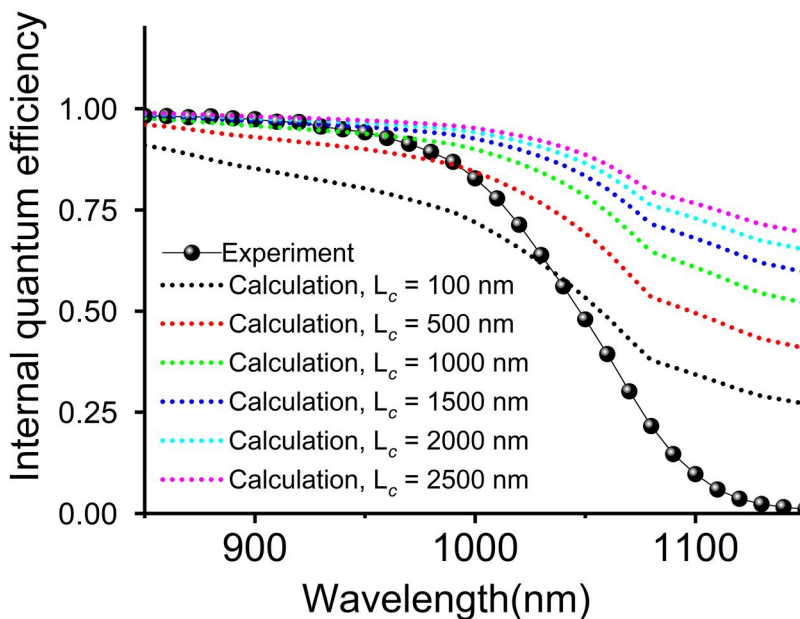
The same framework can also be applied to estimate the CIGS bare front-surface recombination velocity ( $S_f$ ) from the decrease in  $L_{dc}$  observed at short wavelengths. At  $\lambda = 400$ – $600 \text{ nm}$ , where carriers are generated close to the front surface ( $\alpha^{-1} = 27$ – $99 \text{ nm}$ ),  $L_{dc}$  is reduced from the saturated value of  $1811$ – $1815 \text{ nm}$ . Applying the relation  $S_f = 4dD(L_{dc}^{-2} - L_d^{-2})$  with  $d = 1/\alpha$  yields  $S_f$  in the range of approximately  $130$ – $220 \text{ cm/s}$ . This is 2–3 orders of magnitude smaller than the back-surface recombination velocity ( $S_b \sim 5.3 \times 10^4$ – $1.2 \times 10^5 \text{ cm/s}$ ), confirming that carrier losses at the front surface are negligible compared to those at the CIGS/Mo back interface. These values are consistent with reported values, indicating that the front-surface recombination velocity of bare CIGS is smaller than  $10^3 \text{ cm/s}$ ,<sup>25,26</sup> further supporting the validity of the present approach.

To independently cross-validate the minority carrier diffusion length extracted in this work, we applied the relation between internal quantum efficiency (IQE) and the charge-collection length,  $\text{IQE}(\lambda) = 1 - \exp(-\alpha W)/(1 + \alpha L_c)$ .<sup>15</sup> The quantity  $L_c$  is an effective collection length that includes both drift- and diffusion-assisted transport, as well as recombination losses in the

bulk and at the back-surface; it is not a property of the bulk alone. As shown in Fig. 4, no single value of  $L_c$  reproduces the experimentally obtained IQE across the wavelength range examined (850–1150 nm). At the wavelengths of  $\lambda \lesssim 900 \text{ nm}$ , where carrier generation occurs away from the back-surface, the measured IQE coincides with the calculated curve for  $L_c \approx 2 \mu\text{m}$ . This value is in close agreement with the minority carrier diffusion length  $L_d \approx 1.8 \mu\text{m}$  obtained in this work, providing an independent cross-validation of  $L_d$ . At progressively longer wavelengths, where the average generation depth approaches the back-surface, the value of  $L_c$  required to match the measurement decreases monotonically. Within our framework, this decrease is interpreted directly through the deviation of  $L_{dc}(\lambda)$  from its saturated value  $L_d$ .

Among the four recombination pathways in CIGS—front-interface, depletion region, bulk, and back-surface<sup>1</sup>—the present method directly targets the two dominant channels: bulk recombination, reflected in  $L_{db}$  and back-surface recombination, quantified by  $S_b$ . Front-interface recombination is separately reflected in the slight decrease in  $L_{dc}$  at short wavelengths, from which a small front-surface recombination velocity ( $\sim 130$ – $220 \text{ cm/s}$ ) is estimated, while depletion-region recombination is minimized under the short-circuit condition by the strong built-in field. The extracted  $L_d$  and  $S_b$  are, therefore, expected to be only weakly affected by the front-interface and depletion-region channels.

The dominant recombination mechanism in CIGS is known to depend on composition, with quasi-neutral-region recombination dominating in low-Ga absorbers and interface recombination becoming dominant at high Ga.<sup>27</sup> The device demonstrated here employs a standard low-Ga-graded absorber, which lies in the regime where bulk and quasi-neutral-region transport govern the response so that the extraction of  $L_d$  and  $S_b$  is physically meaningful. Extension of the method across the full compositional range—from pure  $\text{CuInSe}_2$  to high-Ga absorbers and various grading



**FIG. 4.** Experimentally obtained IQE (filled circles) of a  $\text{ZnO:Al/i-ZnO/CdS/CIGS/Mo}$  solar cell and IQE calculated from the relation  $\text{IQE}(\lambda) = 1 - \exp(-\alpha W)/(1 + \alpha L_c)$  for  $L_c = 100$ – $2500 \text{ nm}$  (dotted lines). No single  $L_c$  fits the data:  $L_c \approx 2 \mu\text{m}$  reproduces the wavelength range in which carriers are generated sufficiently far from the back-surface, consistent with  $L_d \approx 1.8 \mu\text{m}$  extracted in this work, while progressively smaller  $L_c$  is required at longer wavelengths, where the carrier generation depth approaches the back-surface.

profiles, where interface recombination may become significant—is a meaningful direction for future work.

Having established the extracted values of  $L_d$  and  $S_b$ , we now examine the validity of the diffusion-dominated lateral transport assumption underlying the above analysis. Lateral transport is diffusion-dominated when the lateral drift displacement  $\mu E_x \tau$  is much smaller than the diffusion length  $L_d$ , where  $\mu$  is the minority carrier mobility,  $E_x$  is the lateral component of the electric field, and  $\tau$  is the minority carrier lifetime. Using the Einstein relation  $D = \mu kT/q$  (where  $D$  is the diffusivity,  $k$  is the Boltzmann constant,  $T$  is the temperature, and  $q$  is the elementary charge) and  $L_d = \sqrt{D\tau}$ , this criterion reduces to  $E_x \ll kT/(qL_d)$ , which at room temperature with  $L_d \approx 1.8 \mu\text{m}$  gives  $E_x \ll \sim 140 \text{ V/cm}$ . Three potential sources of lateral electric field were considered. First, compositional non-uniformity produces lateral bandgap fluctuations ( $\sim 20\text{--}50 \text{ meV}$ )<sup>28</sup> that are randomly oriented and are expected to largely average out over the collection length. Second, the electrode-edge field is largely confined to the laterally extended depletion region at the edge, which can be approximated as a quarter-circle of radius  $W = 420 \text{ nm}$ , following the treatment of the narrow-width effect in MOSFETs; since  $W$  is subtracted in extracting  $L_d$ , this region is largely excluded. Third, local band bending from charged defects<sup>29</sup> tends to be randomly distributed or symmetric, so it is unlikely to produce a significant net lateral drift. Taken together, these considerations suggest that the diffusion-dominated assumption is reasonable throughout the analysis.

Note on the treatment of  $\alpha(\lambda)$  is warranted. Throughout this analysis, the absorption coefficient is treated as spatially uniform, although the bandgap of the CIGS layer is graded along the thickness direction. This simplification may introduce uncertainty from depth-dependent variations in  $\alpha$ . Because the framework relies on average generation depth and bounded parameter extraction, the values reported above should be interpreted as effective quantities rather than exact position-resolved parameters. This treatment is consistent with the objective of the present framework, which focuses on extracting bounded and physically meaningful parameters rather than exact spatial profiles.

#### IV. CONCLUSION

A method has been developed to determine the minority carrier diffusion length ( $L_d$ ) and back-surface recombination velocity ( $S_b$ ) in thin-film solar cells using lateral photocurrent measurements. By exploiting the directional asymmetry of carrier transport in bandgap-graded absorbers, the approach isolates the diffusional contribution from mixed drift–diffusion transport in the charge-neutral region. The use of bounded solutions enables the extraction of physically meaningful parameters without requiring detailed knowledge of the bandgap profile or full drift–diffusion modeling.

Application to a  $\text{Cu(In,Ga)Se}_2$  solar cell demonstrates that the method yields values of  $L_d$  and  $S_b$ , consistent with literature, supporting its validity as an analysis framework. While the approach relies on simplifying assumptions, including the use of an effective generation depth, negligible lateral electric fields, and spatially uniform absorption coefficient, these approximations are

physically justified under typical device conditions and allow practical parameter extraction from experimentally accessible data. In particular, the treatment of  $\alpha(\lambda)$  as depth-independent introduces uncertainty in absorbers with strong bandgap grading, and the extracted parameters should accordingly be interpreted as effective values rather than exact spatial quantities.

The proposed method provides a useful tool for analyzing carrier transport and recombination in thin-film photovoltaic devices, particularly in systems where bandgap grading introduces coupled drift–diffusion behavior. It can be extended to other material systems and device architectures where similar transport anisotropy is present. Future work may include numerical simulation to quantify the error bounds and define the applicability range of the analytical framework more precisely.

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#### AUTHOR DECLARATIONS

##### Conflict of Interest

The authors have no conflicts to disclose.

##### Author Contributions

**Eunyeong Yang:** Investigation (equal); Visualization (lead); Writing – original draft (equal). **Jiseong Jang:** Investigation (equal). **Juhyeon Song:** Investigation (supporting). **Jung-Sub Wi:** Investigation (supporting). **Joon Sik Park:** Investigation (supporting). **Sangyeob Lee:** Investigation (equal); Methodology (equal). **Choong-Heui Chung:** Conceptualization (lead); Methodology (equal); Supervision (lead); Writing – original draft (equal); Writing – review & editing (lead).

#### DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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